BDA501 - BIG DATA ANALYTICS

Individual Assignment 3

Land-Cover Classification at Scale with Apache Spark MLlib

Tran Phan Vinh Hoang | 28 September 2026

Abstract

This report extends the Assignment 2 BigEarthNet-S2 pipeline with Apache Spark MLlib. The experiment processes **52,607 patches** from **4 countries** and preserves the official train, validation and test split. Twenty-eight spectral features support multi-class, multi-label and unsupervised learning. Random Forest achieved test accuracy **0.958**, weighted-F1 **0.955** and macro-F1 **0.511**. Binary relevance reached micro-F1 **0.627**. PCA and KMeans identified broad spectral groups, and a reloaded model produced held-out-country accuracy **0.955**. All learned preprocessing is fitted on train only; validation selects models and test is used once.

Study scope

Item	Verified value
Dataset	BigEarthNet-S2 partial HDF5 collection
Scale	52,607 patches; 4 countries
Official split	25,291 train / 13,884 validation / 13,432 test
Feature table	28 numerical features; 0 errors
Tasks	Multi-class, multi-label, PCA, KMeans, scaling and inference

Implemented workflow

Features v2	Multi-class	Multi-label	PCA + KMeans	Deploy
28 features	LR / RF / GBT	19 models	k = 3..10	Save / reload

The full repository was not downloaded. Four HDF5 shards from Assignment 2 meet the minimum scale requirement.

1. Environment, data and experimental design

The verified run uses local Spark/PySpark MLlib on Windows and the required `hdfs_mirror/bigearthnet` layout. This is a local HDFS mirror, not a live distributed HDFS service. Fixed seeds are stored in shared configuration. The official split column is used throughout; randomSplit and random-fold tuning are not used.

Environment item	Verified value
Execution	Apache Spark 3.5.9 / PySpark MLlib, local Windows mode
Storage	hdfs_mirror/bigearthnet
Countries	Ireland (31,330), Lithuania (13,708), Austria (6,911), Finland (658)
Leakage control	Train-fitted transformers; validation selection; final test once
Reproducibility	One script per step, shared config, requirements and seeds

2. Step 1 - richer feature engineering

The executor-side job retains the Assignment 2 features and adds mean and standard deviation for B05, B06, B07, B8A, B11 and B12. These bands remain on their native 60 x 60 grid. NDMI and NDBI are computed per pixel and averaged; NDVI p10, p50 and p90 preserve within-patch variation.

Feature group	Columns	Count
Assignment 2 base	B02/B03/B04/B08 mean and std; NDVI; NDWI; brightness	11
20 m bands	B05/B06/B07/B8A/B11/B12 mean and std	12
New indices	NDMI and NDBI	2
NDVI distribution	p10, p50 and p90	3
Total	Saved to bigearthnet/features_v2	28

2.1 Validation

Check	Result
Input / output rows	52,607 / 52,607
Extraction errors	0
Null / NaN values	0 / 0
Index range failures	0
Runtime	82.077 s

Row preservation, zero errors and bounded indices show that richer features were added without invalid numerical values.

3. Step 2 - multi-class classification

The task keeps patches containing exactly one selected frequent label. The partial data contains 18 observed labels, but only 13 appear as exact-one classes in train. Several classes have fewer than ten rows, so inverse-frequency weights and macro-F1 are essential.

3.1 Exact-one class counts by official split

Exact-one target	Train	Validation	Test
Pastures	1714	947	738
Arable land	442	353	363
Marine waters	4966	819	103
Land principally occupied by agriculture, with significant areas of natural vegetation	1	0	0
Coniferous forest	10	6	5
Mixed forest	3	0	6
Broad-leaved forest	64	78	4
Complex cultivation patterns	0	0	2
Urban fabric	33	4	0
Inland wetlands	86	57	13
Inland waters	89	19	0
Permanent crops	3	0	0
Moors, heathland and sclerophyllous vegetation	6	0	3
Natural grassland and sparsely vegetated areas	4	0	0

3.2 Leakage-safe pipeline

StringIndexer	VectorAssembler	StandardScaler	Classifier
Fit on train	28 inputs	Fit on train	LR / RF / GBT-OvR

Class weights come from train only. Each ParamGridBuilder configuration is fitted on train and selected by validation macro-F1.

3.3 Model selection and validation

Model	Configuration	Val. macro-F1	Time (s)
LR	regParam=0.0, elasticNetParam=0.0	0.735	20.537
LR	regParam=0.0, elasticNetParam=0.5	0.735	14.593
LR	regParam=0.01, elasticNetParam=0.0	0.684	12.204
LR	regParam=0.01, elasticNetParam=0.5	0.623	15.317
RF	numTrees=20, maxDepth=5	0.797	6.141
RF	numTrees=20, maxDepth=8	0.828	6.473
RF	numTrees=40, maxDepth=5	0.814	4.431
RF	numTrees=40, maxDepth=8	0.845	5.544
GBT-OvR	maxIter=5, maxDepth=3	0.715	54.915
GBT-OvR	maxIter=10, maxDepth=3	0.767	83.295

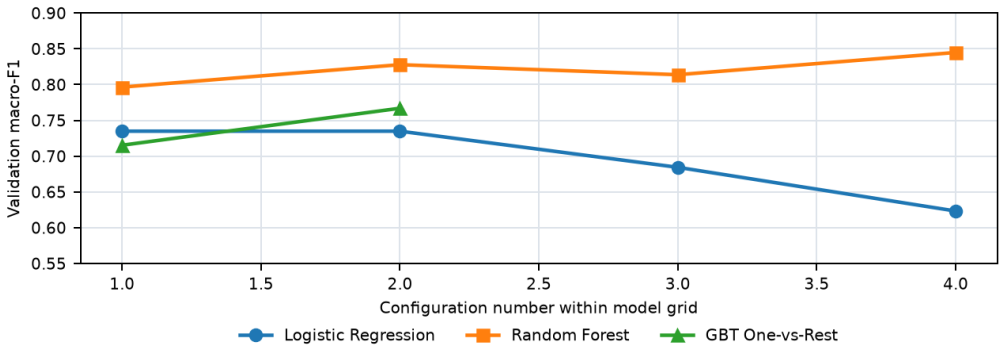


Figure 1. Validation macro-F1 for every configuration. RF with 40 trees and depth 8 was selected at 0.845. GBT-OneVsRest was much slower locally.

3.4 Test performance and Assignment 2 baselines

The selected RF was evaluated once on 1,237 test rows: accuracy **0.958**, weighted-F1 **0.955**, and macro-F1 **0.511**. The large accuracy-to-macro-F1 gap exposes weak long-tail performance.

Model	Accuracy	Macro-F1	Scope
A2 majority	0.554	0.238	A2 three-class
A2 spectral rule	0.817	0.858	A2 three-class
A2 centroid - 11	0.800	0.837	A2 three-class
A2 centroid - 5	0.809	0.853	A2 three-class
A3 Random Forest	0.958	0.511	A3 exact-one

Assignment 2 baselines use only Pastures, Arable land and Marine waters; the A3 RF has a broader label space. The table provides continuity, not a direct ranking.

3.5 Per-class test behaviour

Test class	Support	Precision	Recall	F1
Pastures	738	0.957	0.999	0.977
Marine waters	103	1.000	1.000	1.000
Arable land	363	0.985	0.920	0.952
Coniferous forest	5	0.429	0.600	0.500
Inland wetlands	13	0.385	0.385	0.385
Unseen-in-train bucket	2	0.000	0.000	0.000
Mixed forest	6	1.000	0.167	0.286
Broad-leaved forest	4	0.500	0.500	0.500
Moors, heathland and sclerophyllous vegetation	3	0.000	0.000	0.000

3.6 Interpretation and top-k retraining

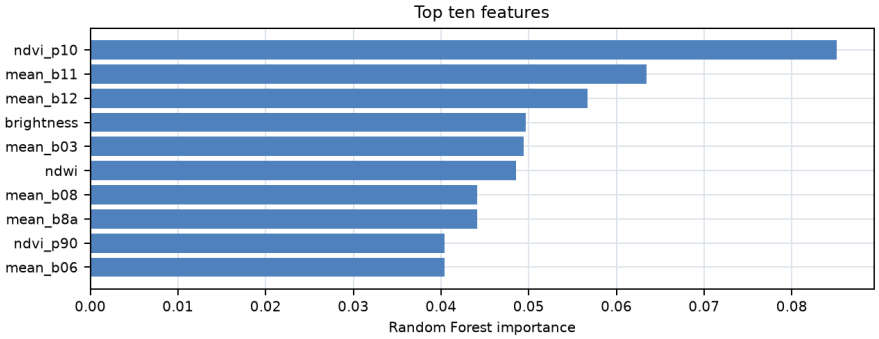


Figure 2. NDVI p10 ranks first; SWIR means B11/B12 and visible/NIR summaries also contribute. LR coefficients rank std_b05, mean_b12, std_b11 and NDVI p10 highest.

Features	Accuracy	Weighted-F1	Macro-F1
Full 28	0.958	0.955	0.511
Top 5	0.793	0.861	0.439
Top 10	0.815	0.881	0.443

Top-5 and top-10 models reduce macro-F1, showing that rare classes need information across the full feature set.

4. Step 3 - multi-label classification

Nineteen binary columns are created. A train-fitted feature PipelineModel is reused by one Logistic Regression classifier per class. Thresholds from 0.1 to 0.9 are selected on validation and then applied once to test.

Metric	Value	Meaning
Micro-precision	0.533	All decisions pooled
Micro-recall	0.761	All decisions pooled
Micro-F1	0.627	Frequent labels dominate
Macro-F1	0.397	Equal class weight

4.1 Per-class multi-label performance

Class	Train +	Threshold	Precision	Recall	F1
Urban fabric	3,822	0.7	0.405	0.353	0.377
Industrial or commercial units	732	0.9	0.294	0.184	0.226
Arable land	11,402	0.4	0.763	0.896	0.824
Permanent crops	1,027	0.9	0.281	0.253	0.266
Pastures	12,504	0.4	0.859	0.898	0.878
Complex cultivation patterns	3,858	0.6	0.409	0.587	0.482
Land principally occupied by agriculture, with significant areas of natural vegetation	5,020	0.5	0.354	0.789	0.489
Agro-forestry areas	0	N/A	0.000	0.000	0.000
Broad-leaved forest	3,970	0.5	0.296	0.806	0.433
Coniferous forest	4,702	0.7	0.676	0.633	0.654
Mixed forest	4,507	0.6	0.477	0.699	0.567
Natural grassland and sparsely vegetated areas	316	0.7	0.084	0.290	0.130
Moors, heathland and sclerophyllous vegetation	876	0.8	0.291	0.257	0.273
Transitional woodland, shrub	3,162	0.6	0.408	0.726	0.522
Beaches, dunes, sands	242	0.9	0.000	0.000	0.000
Inland wetlands	2,023	0.7	0.188	0.456	0.266
Coastal wetlands	50	0.5	0.000	0.000	0.000
Inland waters	1,359	0.7	0.180	0.443	0.256
Marine waters	5,663	0.8	0.853	0.957	0.902

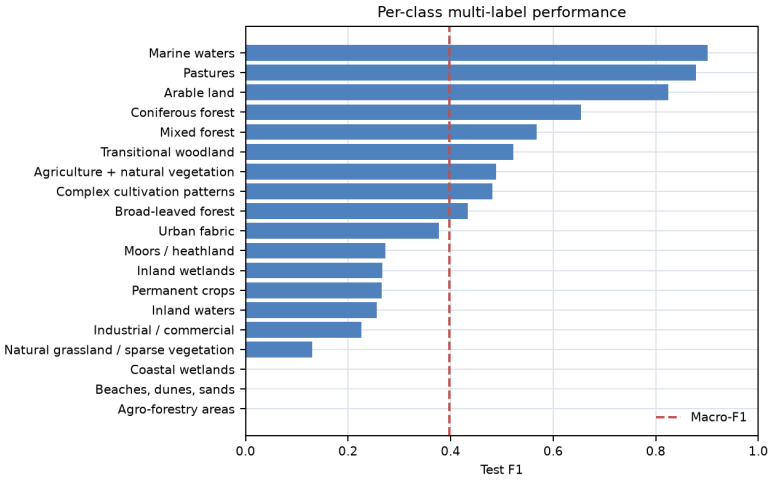


Figure 3. Test F1 for all 19 labels. Marine waters, Pastures and Arable land are strongest. Agro-forestry has no positive training rows; Coastal wetlands and Beaches/dunes/sands are too rare to produce true positives.

5. Step 4 - unsupervised analysis

StandardScaler and PCA are fitted on train inside a Pipeline. PC1 explains **72.1%** and PC2 **12.3%**; together they retain **84.4%** of variance.

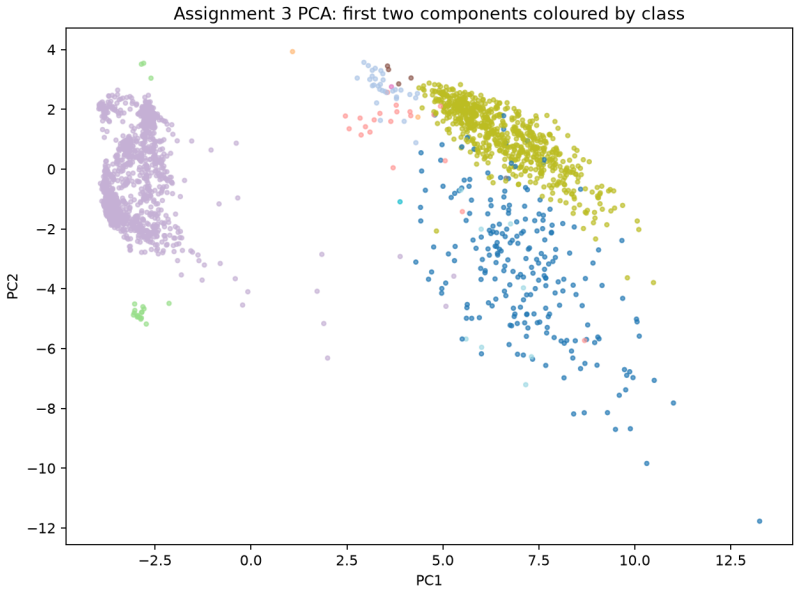


Figure 4. PC1 and PC2 coloured by exact-one class. Marine waters separates clearly; vegetation and agriculture overlap because summaries remove spatial texture.

5.1 KMeans model selection

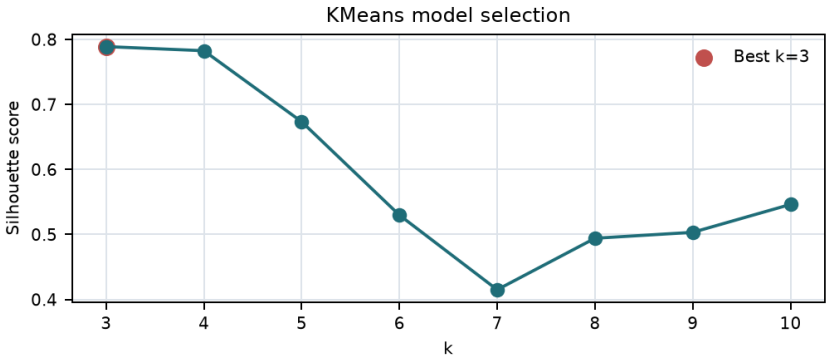


Figure 5. Silhouette for k=3..10. The maximum is 0.789 at k=3.

Cluster	Dominant class	Count	Interpretation
0	Marine waters	5,841	Broad spectral regime
1	Arable land	1,061	Broad spectral regime
2	Pastures	3,340	Broad spectral regime

Clusters do not equal semantic classes. Cluster 0 is mostly Marine waters, cluster 1 includes Arable land, and cluster 2 is mostly Pastures. Multi-label semantics and Euclidean distance prevent a perfect match.

6. Step 5 - scalability and deployment

The selected RF is retrained on stratified 25%, 50% and 100% samples with four and eight partitions. The sampled training frame is cached. Timings describe this laptop and are not a claim of linear cluster scaling.

Train fraction	Partitions	Rows	Time (s)	Test macro-F1
25%	4	1,567	8.763	0.507
25%	8	1,567	5.010	0.515
50%	4	3,240	5.793	0.547
50%	8	3,240	5.261	0.538
100%	4	6,567	4.139	0.626
100%	8	6,567	5.420	0.585

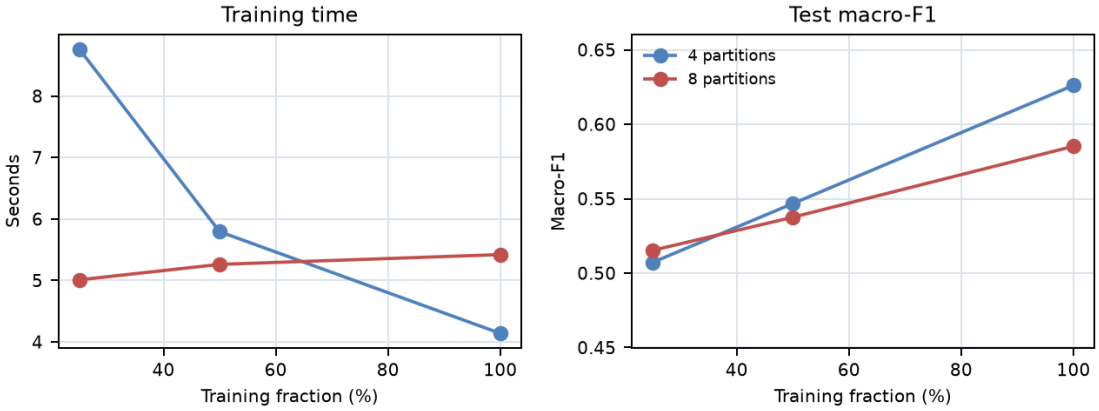


Figure 6. Training time and test macro-F1. Full data gives the best macro-F1; partition effects vary because scheduling overhead matters locally.

6.1 Persistence

The PipelineModel is saved under `bigearthnet/models`, reloaded and evaluated again. Exact metric reproduction: **True**.

Reloaded model	Rows	Accuracy	Weighted-F1	Macro-F1
RF PipelineModel	1,197	0.966	0.962	0.574

6.2 Batch inference and Domain shift

Held-out country	Rows	Accuracy	Weighted-F1	Macro-F1
Lithuania	956	0.955	0.952	0.513

Lithuania is excluded from fitting and used for country-level inference. Accuracy remains high because common classes dominate, but macro-F1 falls to 0.513. This indicates domain shift in class mixture and spectral conditions. Predictions are saved under `bigearthnet/predictions` partitioned by country.

7. Discussion

The project provides a reproducible Spark MLlib workflow across supervised, multi-label, unsupervised and deployment tasks. NDVI percentiles and SWIR features rank highly because they describe vegetation variation, moisture and built surfaces. Accuracy must be read with macro-F1: frequent classes are learned well, while rare classes have little support and may be absent from a split.

The multi-label results show the same pattern. Patch means, standard deviations and percentiles remove texture, edges, shapes and spatial arrangement. An image model could use these signals but would require more compute and a different scope.

7.1 Limitations and transparent deviations

Limitation	Effect and treatment
Partial dataset	52,607 patches meet scale, but the complete corpus was not downloaded.
18 of 19 labels	Agro-forestry has no positive examples and cannot be learned.
Fewer than 35 exact-one classes	Only 13 occur in train; scripts request 35 and report availability.
Local Spark / HDFS mirror	No claim of cluster speed-up or live HDFS evidence.
Patch summaries	Spectral statistics do not represent texture or geometry.

7.2 Reproducibility

Run the scripts in order; all use shared configuration from `scripts/a3_config.py`:

```
.\.venv\Scripts\python.exe scripts\05_a3_features_v2.py --partitions 16
.\.venv\Scripts\python.exe scripts\06_a3_multiclass.py
.\.venv\Scripts\python.exe scripts\07_a3_multilabel.py
.\.venv\Scripts\python.exe scripts\08_a3_unsupervised.py
.\.venv\Scripts\python.exe scripts\09_a3_scalability_deploy.py
```

8. Conclusion

All five required stages were completed on 52,607 patches with the official split. RF reached 0.958 accuracy and 0.511 macro-F1; multi-label micro-F1 was 0.627; KMeans selected k=3; and the saved model reproduced its metrics before held-out-country scoring. The main constraints are partial-data class coverage and patch-level representation.

AI assistance statement

OpenAI Codex was used to help implement, verify and present this assignment. All counts, timings and metrics come from submitted JSON evidence and can be reproduced from the scripts.